SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PETROCHEMICAL TRADE ANALYTICS & PREDICTIVE INTELLIGENCE

Standard / Classification:	IEEE Std 830-1998 / ISO/IEC/IEEE 29148 Standard SRS
Evaluation Body:	HCL Career Shaper COE & NASSCOM FutureSkills Prime
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Target Horizon:	2015 – 2024 UN Comtrade Benchmark ($302.18B Gross Trade)

1. Introduction & Product Scope

1.1 Purpose: This document defines the formal software requirements specification for the 'Petrochemical Trade Analytics & Predictive Intelligence' platform in accordance with IEEE Std 830-1998 standards, serving as the official technical contract for the HCL Career Shaper COE and NASSCOM capstone assessment.
1.2 Scope: The software platform automates the end-to-end ingestion, econometric analysis, and price forecasting of global petrochemical commodity trade across eight reporter economies (China, Germany, India, Japan, Netherlands, South Korea, Saudi Arabia, USA) and five building-block feedstocks: Ethylene (HS 290121), Propylene (HS 290122), Butadiene (HS 290124), Benzene (HS 290220), and Toluene (HS 290230). Cumulative trade evaluated totals 334.13 Mt volume and $302.18 Billion USD.
1.3 Intended Audience: Academic evaluators, enterprise trade analysts, and software engineers.

2. Overall Description & System Architecture

2.1 Product Perspective: A modular 5-tier architecture integrating:
• Data Engineering & ETL: Schema normalization, unit harmonization (TUV = USD/MT), and temporal lag generation.
• Relational Warehousing: Production SQLite 3NF Star Schema (Fact_TradeTransactions linked to Dim_Date, Dim_Country, Dim_Commodity).
• Statistical Inference: Welch's t-test (t = -4.858, p = 1.42e-6) and One-way ANOVA (F = 11.23, p = 2.70e-9).
• Machine Learning: Gradient Boosting (R^2 = 0.9543), Random Forest (R^2 = 0.9471), and K-Means corridor clustering.
• Reactive UI: Multi-page Streamlit web app with dual Dark/Light mode theming, global Choropleth map, and Sankey corridors.
2.2 Operating Environment: Python 3.13, Windows 10/11 or Linux, SQLite3, Chromium Web Browsers, < 16 MB artifact limit.

3. Functional Requirements (IEEE 830)

[FR-01] Data Ingestion & Sanitization: Automated ingestion of UN Comtrade transactional CSVs with missing quantity imputation.
[FR-02] Statistical Testing Suite: Real-time parametric evaluation of market structural shocks and inter-commodity price dispersion.
[FR-03] Predictive Regression Engine: Multi-model predictive pipeline with hyperparameter tuning achieving R^2 >= 0.95.
[FR-04] Star Schema SQL Analytics: Analytical query execution supporting CTEs, ROW_NUMBER(), LAG(), and DENSE_RANK().
[FR-05] Interactive Web UI: Dynamic Choropleth geospatial trade map, bilateral Sankey cargo diagrams, and What-If scenario simulation.

4. Non-Functional Requirements (NFRs)

• **Performance:** UI re-render latency < 1.5 seconds; local SQLite query latency < 20 ms; ML inference latency < 15 ms.
• **Reproducibility:** Deterministic execution with fixed random seeds (random_state=42) across all ML models.
• **Usability:** One-click batch file execution (run.bat) without complex manual environment setup.
• **File Size Compliance:** All generated PDFs, DOCX files, and datasets strictly remain below 16 MB.

5. Verification & Acceptance Criteria

The software successfully satisfies all HCL Career Shaper COE and NASSCOM submission mandates: complete compiled source code, photographic evidence of the deployed system, formal project synopsis, verified datasets, and econometric validation results.